Anti-inflammatory potential of a lipolotion containing coriander oil in the ultraviolet erythema test.

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SUMMARY BACKGROUND: Coriander oil is used as an antimicrobial agent and as a natural fragrance. The present study investigated the anti-inflammatory potency of coriander oil in the ultraviolet (UV) erythema test in vivo. METHODS: 40 volunteers were enrolled in this monocentric,randomized,placebo-controlled double-blind study.Test areas on the back were irradiated with the 1.5 fold minimal erythema dose UV-B. Subsequently, the test areas were treated under occlusion for 47 hours with a lipolotion containing 0.5% or 1.0% essential coriander oil. Hydrocortisone (1.0%) and betamethasone valerate (0.1%) in the vehicle served as positive controls.The vehicle was used as place-bo.The effect of the test substances on the UV-induced erythema was measured photometrically after 48 hours.Additionally,the skin tolerance of the test preparations was assessed on non-irradiated skin. RESULTS: Compared to placebo, the lipolotion with 0.5% coriander oil significantly reduced the UV-induced erythema, but it was not as effective as hydrocortisone. The skin tolerance of both coriander oil concentrations was excellent. CONCLUSIONS: The lipolotion containing coriander oil displayed a mild antiinflammatory effect in this study. It could be useful in the concomitant treatment of inflammatory skin diseases.